

MTH203: Sequences and series

Lecture 14

In the previous lecture, we proved the Cauchy Criterion for convergence of series, which we quote here as follows (see Lecture 13, Theorem 8):

Theorem 1 (Cauchy Criterion for convergence of series). *Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers. The following two statements are equivalent:*

(a) *For every $\epsilon > 0$, there exists an integer N such that for any integers m and n satisfying $n > m > N$, we have*

$$|a_{m+1} + \cdots + a_n| < \epsilon.$$

(b) *The series $\sum_{n=1}^{\infty} a_n$ converges.*

Observe that this gives a *necessary and sufficient* condition. Thus, not only can we use it to prove that a series is convergent, it also gives us some information about the behaviour of the terms of a series that is *known* to be convergent. An important application of this result is the Absolute Convergence Test.

1. ABSOLUTE CONVERGENCE TEST

We now look at a very useful criterion for proving convergence.

Theorem 2 (Absolute Convergence Test). *Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers. Suppose that the series $\sum_{n=1}^{\infty} |a_n|$ converges. Then, the series $\sum_{n \in \mathbb{N}} a_n$ also converges.*

Proof. By the Cauchy Criterion for the convergence of series (see Lecture 13, Theorem 8), for any $\epsilon > 0$, there exists an integer N such that if $n > m > N$, we have

$$|a_{m+1}| + \cdots + |a_n| < \epsilon.$$

By the triangle inequality, we have

$$|a_{m+1} + \cdots + a_n| \leq |a_{m+1}| + \cdots + |a_n| < \epsilon.$$

Thus, by applying the Cauchy Criterion to the series $\sum_{n=1}^{\infty} a_n$, we see that this series converges. $\square$

Note that this is *not* a necessary and sufficient criterion for convergence. This is only a *sufficient* condition for convergence, but it is not necessary. In other words, it is very much possible that a particular series $\sum_{n=1}^{\infty} a_n$ may converge even if its “absolute version”, the series $\sum_{n=1}^{\infty} |a_n|$, does not converge.

Definition 3. A series $\sum_{n=1}^{\infty} a_n$ is said to *converge absolutely* if the series $\sum_{n=1}^{\infty} |a_n|$ converges.

If a series $\sum_{n=1}^{\infty} a_n$ converges, but the series $\sum_{n=1}^{\infty} |a_n|$ does not converge, we say that $\sum_{n=1}^{\infty} a_n$ *converges conditionally*.

Example 4. The series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is convergent. (We will prove this later.) However, the series $\sum_{n=1}^{\infty} \frac{1}{n}$ is not convergent. (We saw this in Lecture 11, Example 8.) Thus, the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is conditionally convergent.

2. RATIO TEST

The Ratio Test for convergence is a criterion that is obtained by combining the Absolute Convergence Test, the Comparison Test and our understanding of the convergence of geometric series.

Recall that we have proved that the series $\sum_{k=0}^{\infty} r^k$ converges if and only if $|r| < 1$. It follows that for any non-zero real number c , the series $\sum_{k=0}^{\infty} cr^k$ converges if and only if $|r| < 1$. (The series converges trivially if $c = 0$, regardless the value of $|r|$.)

Theorem 5. Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers. Suppose there exists an integer N and a real number $0 \leq r < 1$ such that for every $n > N$, we have $|a_{n+1}/a_n| \leq r$. Then, the series $\sum_{n=1}^{\infty} a_n$ is convergent.

Proof. Let N and r be as in the statement of the theorem. Then, we see that $|a_{N+2}/a_{N+1}| \leq r$. Thus, $|a_{N+2}| \leq |a_{N+1}|r$.

We claim that for any integer $k \geq 1$, we have $|a_{N+k+1}| \leq |a_{N+1}|r^k$. This has already been verified for $k = 1$.

Suppose that for some integer $k \geq 1$, it is known that $|a_{N+k+1}| \leq |a_{N+1}|r^k$. Then, as $|a_{N+k+2}/a_{N+k+1}| \leq r$, we have

$$|a_{N+k+2}| \leq |a_{N+k+1}|r \leq |a_{N+1}|r^k \cdot r = |a_{N+1}|r^{k+1}.$$

This completes the induction step and proves our claim for any integer k .

Now, we consider the two series $\sum_{k=0}^{\infty} |a_{N+k+1}|$ and $\sum_{k=0}^{\infty} |a_{N+1}|r^k$. We have seen that for every k , we have $0 \leq |a_{N+k+1}| \leq |a_{N+1}|r^k$. The series $\sum_{k=0}^{\infty} |a_{N+1}|r^k$ converges since $|r| < 1$. Thus, by the Comparison Test (Lecture 12, Theorem 6), we see that the series $\sum_{k=0}^{\infty} |a_{N+k+1}|$ converges.

By the Absolute Convergence Test proved above (see Theorem 2, it follows that the series $\sum_{k=0}^{\infty} a_{N+k+1}$ also converges. This series is obtained by just dropping the first N terms from the series $\sum_{n=1}^{\infty} a_n$. Thus, the series $\sum_{n=1}^{\infty} a_n$ also converges. (See Lecture 12, Proposition 2.) $\square$

Remark 6. In the statement of the above theorem, we have $|a_{n+1}/a_n| \leq r$ for all $n > N$. Since $r < 1$, this means $|a_{n+1}/a_n| < 1$ for all $n > N$. However, the statement of the theorem cannot be simplified by just saying that $|a_{n+1}/a_n|$ should be less than 1 for all $n > N$. Indeed, consider the series $\sum_{n=1}^{\infty} (1/n)$. Then $\frac{1/(n+1)}{1/n} = n/(n+1) < 1$ for all n . However, the series $\sum_{n=1}^{\infty} (1/n)$ diverges. The reason is that even though $1/(n+1)1/n < 1$ for all n , we have $\lim_{n \rightarrow \infty} 1/(n+1)1/n = 1$. In other words, the ratios of the successive terms slowly increase and get arbitrarily close to 1. Thus, we cannot compare this series with any series of the form $\sum_{n=0}^{\infty} ar^n$ for any $r < 1$ and thus the Comparison Test cannot be applied.

The moral of the story is that to deduce the convergence of the series using the Ratio Test, the absolute values of the ratios of successive terms $|a_{n+1}/a_n|$ should not just be less than one, but need to stay *away* from 1.

The following corollary is also often referred to as the Ratio Test:

Corollary 7. Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers such that $\lim_{n \rightarrow \infty} |a_{n+1}/a_n|$ exists and is less than 1. Then the series $\sum_{n=1}^{\infty} a_n$ converges.

Proof. Let $L = \lim_{n \rightarrow \infty} |a_{n+1}/a_n|$. Since $L < 1$, $1 - L > 0$. Let ϵ be any real number such that $0 < \epsilon < 1 - L$. (For example, we may take $\epsilon = (1 - L)/2$.) Then, it follows that $L + \epsilon < 1$.

As $\lim_{n \rightarrow \infty} |a_{n+1}/a_n| = L$, there exists an integer N such that $||a_{n+1}/a_n| - L| < \epsilon$. Thus, $|a_{n+1}/a_n| - L < \epsilon$ and hence $|a_{n+1}/a_n| < L + \epsilon < 1$. We may now apply Theorem 5 with $r = L + \epsilon$ to conclude that the series $\sum_{n=1}^{\infty} a_n$ converges. $\square$

3. ROOT TEST

The Root Test is very similar to the Ratio Test and the proof is also based on a similar idea.

Theorem 8. Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers. Suppose there exists an integer N and a real number $0 \leq r < 1$ such that for every $n > N$, we have $\sqrt[n]{|a_n|} \leq r$. Then, the series $\sum_{n=1}^{\infty} a_n$ is convergent.

Note that $\sqrt[n]{|a_n|}$ denotes the unique *non-negative* real number α such that $\alpha^n = |a_n|$. The existence of such n -th roots can be proved in a manner very similar to the proof of the existence of a square root of 2. (See Lecture 4, Example 6.)

Proof. Let N and r be as in the statement of the theorem. Then, for any $n > N$, we have $\sqrt[n]{|a_n|} \leq r$. Thus, $|a_n| \leq r^n$.

Now, we consider the two series $\sum_{k=N+1}^{\infty} |a_k|$ and $\sum_{k=N+1}^{\infty} r^k$. We have seen that for every $k > N$, we have $0 \leq |a_k| \leq r^k$. The series $\sum_{k=N}^{\infty} r^k$ is a geometric series and since $r < 1$, it converges. Thus, by the Comparison Test (Lecture 12, Theorem 6), we see that the series $\sum_{k=N}^{\infty} |a_k|$ converges.

By the Absolute Convergence Test proved above (see Theorem 2, it follows that the series $\sum_{k=N+1}^{\infty} a_k$ also converges. This series is obtained by just dropping the first N terms from the series $\sum_{n=1}^{\infty} a_n$. Thus, the series $\sum_{n=1}^{\infty} a_n$ also converges. (See Lecture 12, Proposition 2.) $\square$

4. REMARKS ABOUT THE RATIO TEST AND THE ROOT TEST

Firstly, we note that both the Ratio Test as well as the Root Test only provide sufficient conditions for convergence of series. Even if the conditions given in the statement of the theorem are not true, it is quite possible that the series may converge. We will look at this in greater detail in the next lecture.

Let us look at two examples where the Root Test works, but the Ratio Test fails.

Example 9. Consider the series $(a_n)_{n \in \mathbb{N}}$ where

$$a_n = \begin{cases} 0 & \text{if } n \text{ is odd,} \\ 1/2^{n/2} & \text{if } n \text{ is even.} \end{cases}$$

Then

$$\sqrt[n]{|a_n|} = \begin{cases} 0 & \text{if } n \text{ is odd,} \\ 1/\sqrt{2} & \text{if } n \text{ is even.} \end{cases}$$

Thus, the Root Test allows us to deduce that the series converges.

On the other hand, the Ratio Test cannot even be applied since $|a_{n+1}/a_n|$ is not even defined if n is odd.

Thus, we see if there are infinitely many zeros in the sequence $(a_n)_{n \in \mathbb{N}}$, we cannot apply the Ratio Test to the series $\sum_{n=1}^{\infty} a_n$. However, this is not the only way this test might fail to work, as we will see in the next example.

Example 10. Consider the series $(a_n)_{n \in \mathbb{N}}$ where

$$a_n := \begin{cases} 1/2^{n+1} & \text{if } n \text{ is odd,} \\ 1/2^n & \text{if } n \text{ is even.} \end{cases}$$

Then

$$\sqrt[n]{|a_n|} = \begin{cases} \sqrt[n]{1/2} \cdot (1/2) & \text{if } n \text{ is odd} \\ 1/2 & \text{if } n \text{ is even.} \end{cases}$$

For any n , we have $\sqrt[n]{1/2} < 1$ and so when n is odd, we have $\sqrt[n]{|a_n|} < 1/2$. Thus, the Root Test tells us that the series converges.

On the other hand,

$$|a_{n+1}/a_n| = \begin{cases} 1 & \text{if } n \text{ is odd,} \\ 1/4 & \text{if } n \text{ is even.} \end{cases}$$

Thus, the Ratio Test cannot be applied.

REFERENCES

- [1] Stephen Abbott, *Understanding Analysis*, Undergraduate Texts in Mathematics, Springer, 2015.